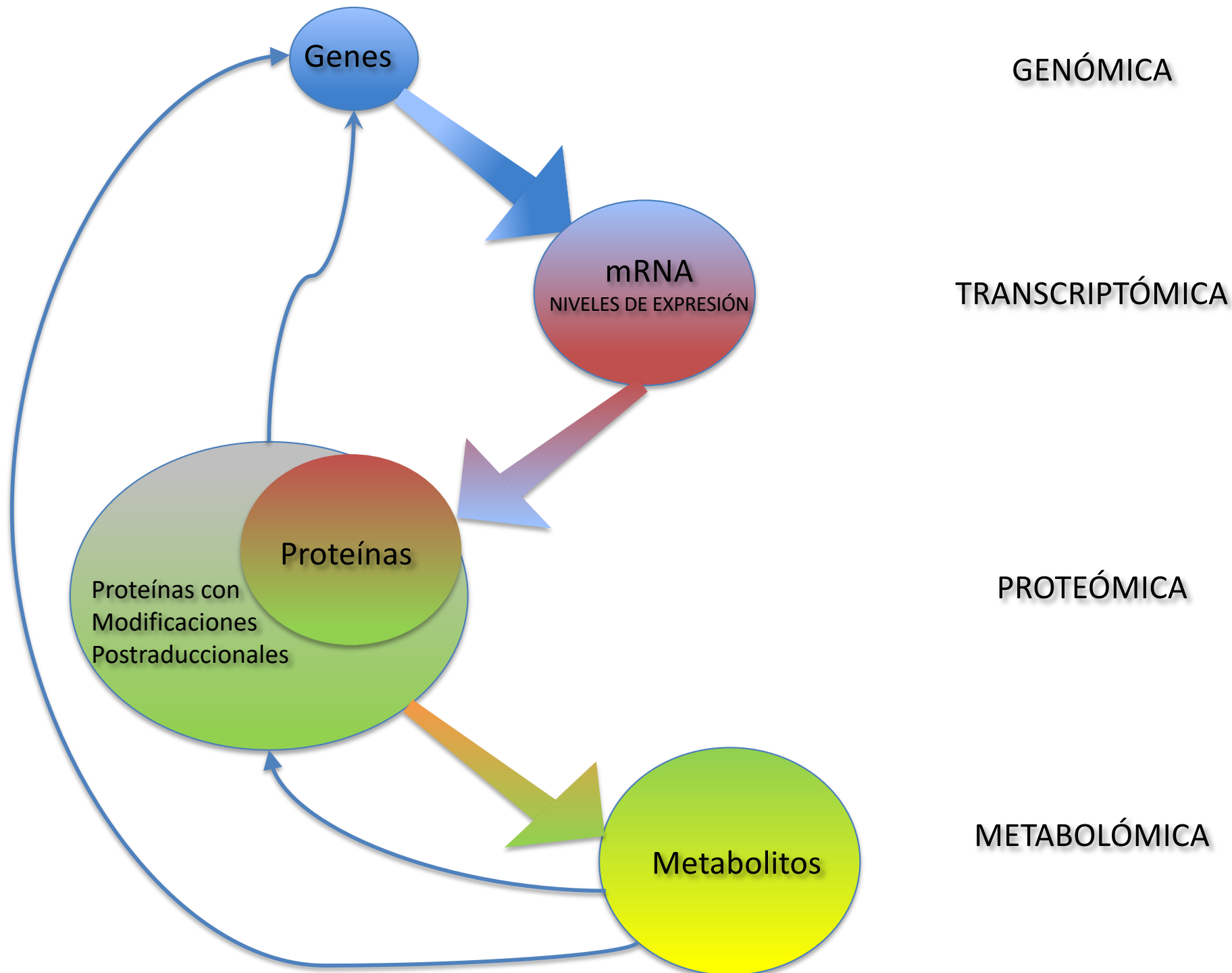
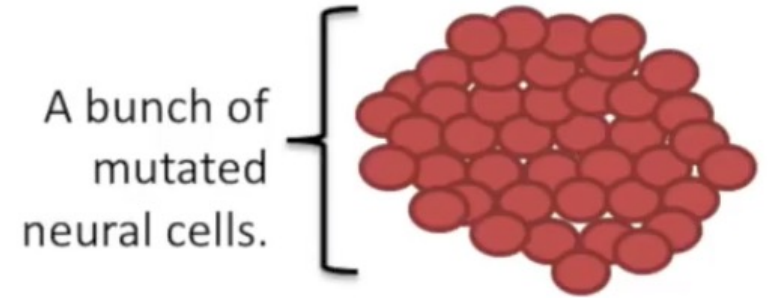
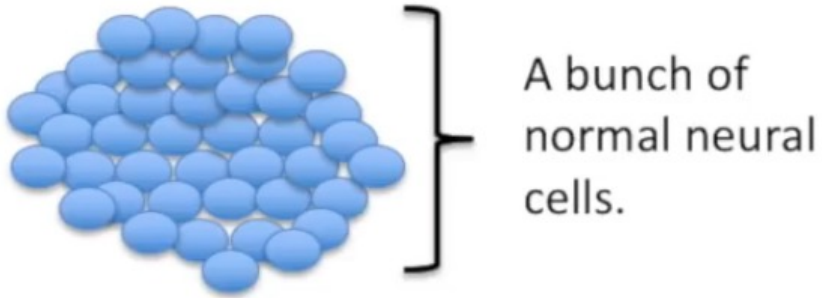




RNAseq

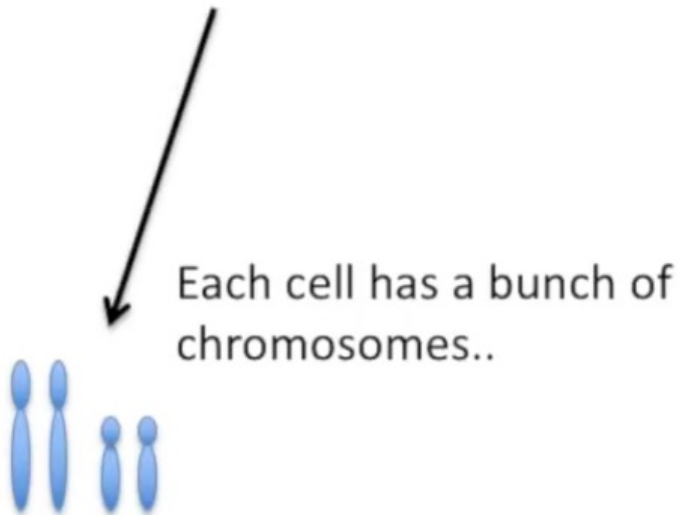
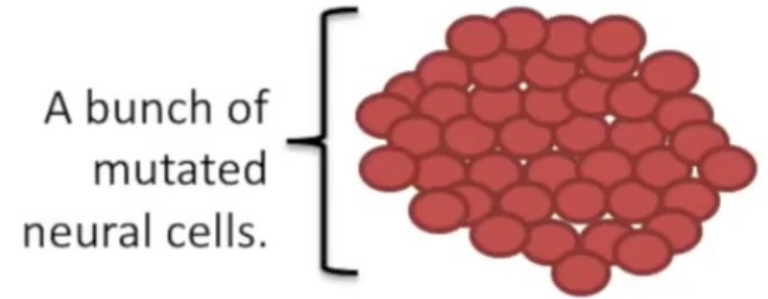
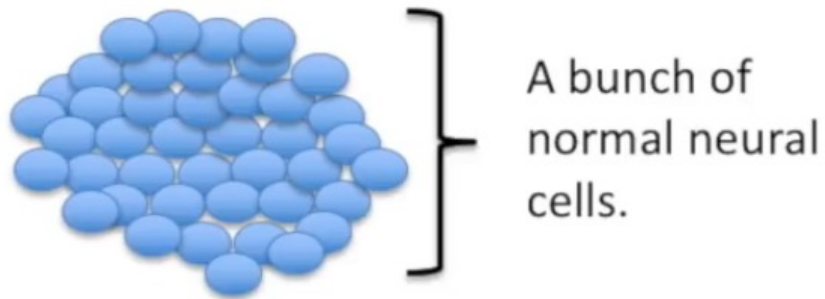
 = a normal neural cell

 = a mutated neural cell



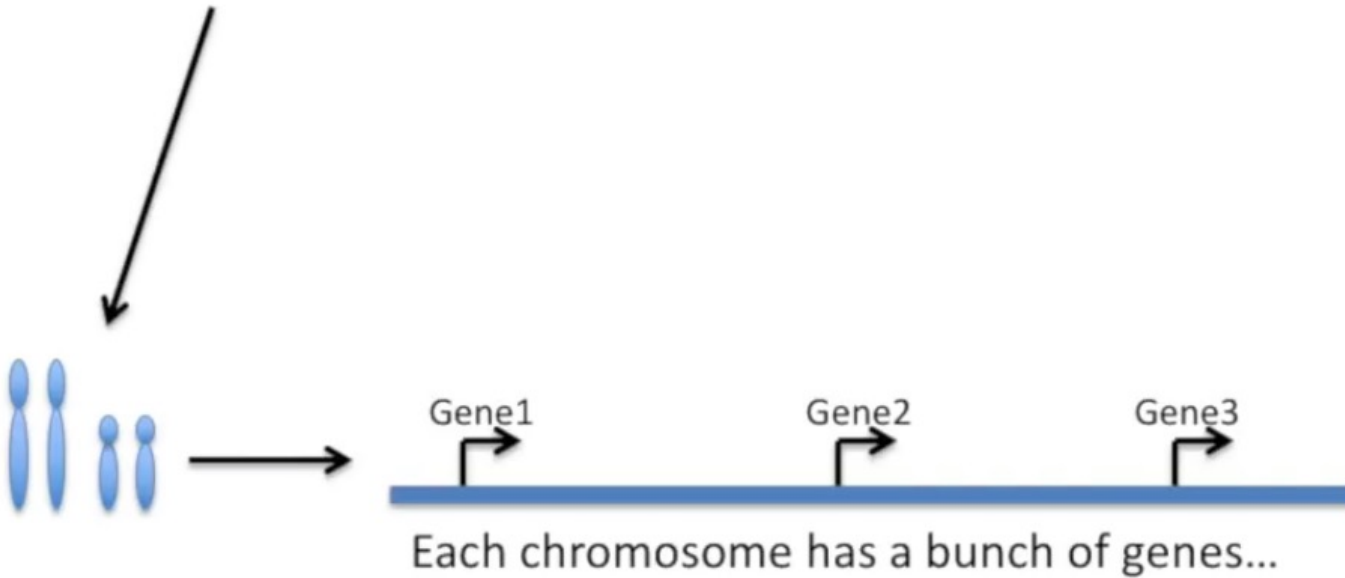
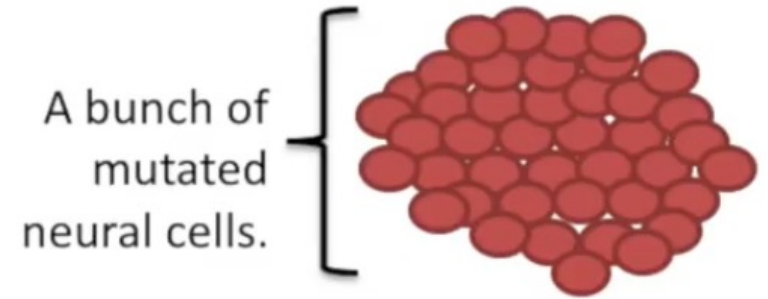
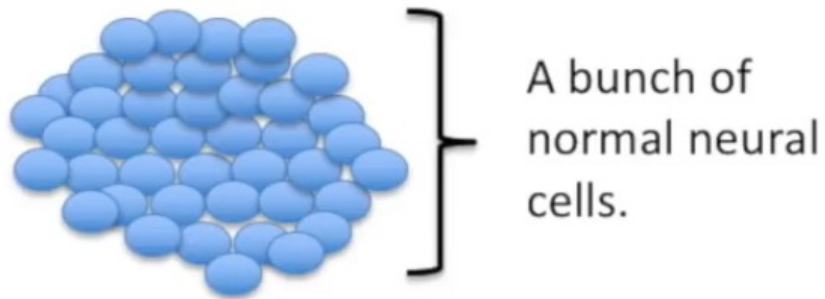
 = a normal neural cell

 = a mutated neural cell



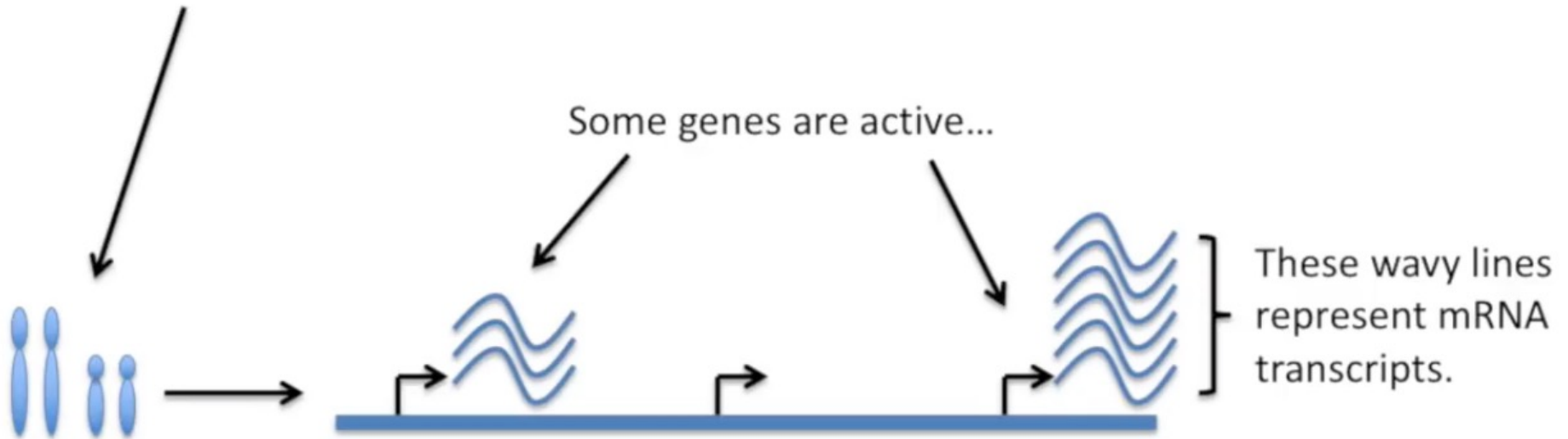
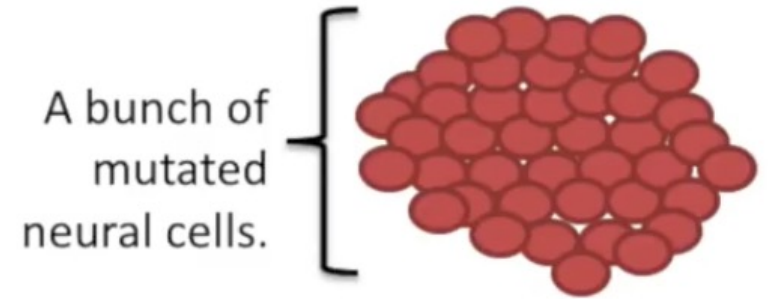
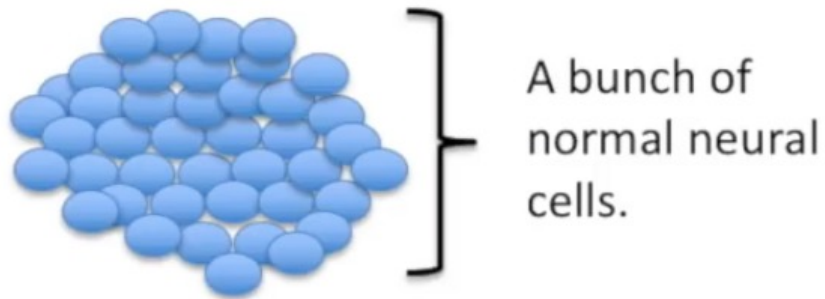
● = a normal neural cell

● = a mutated neural cell



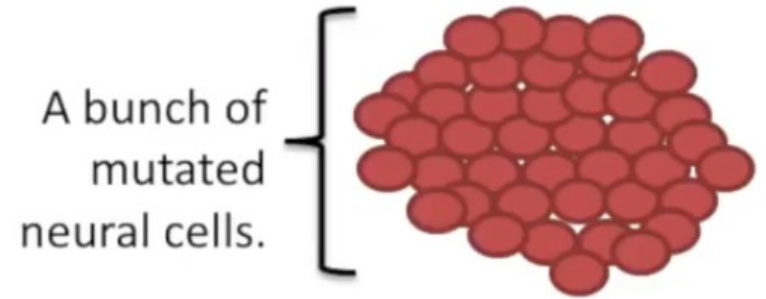
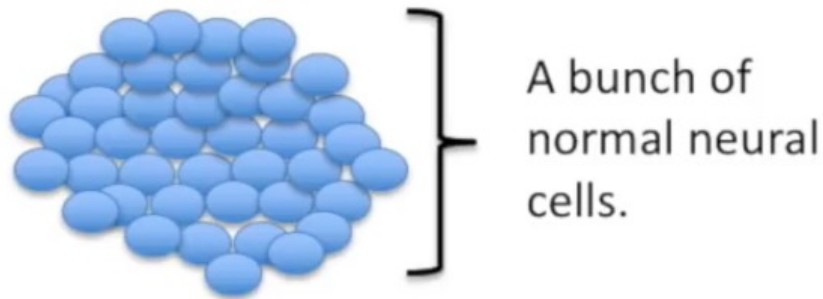
● = a normal neural cell

● = a mutated neural cell

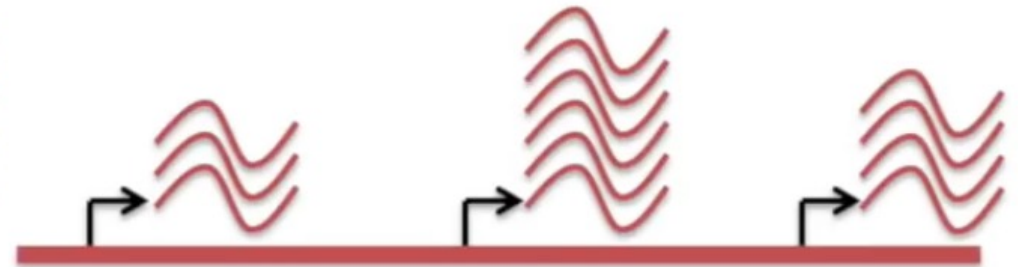


● = a normal neural cell

● = a mutated neural cell



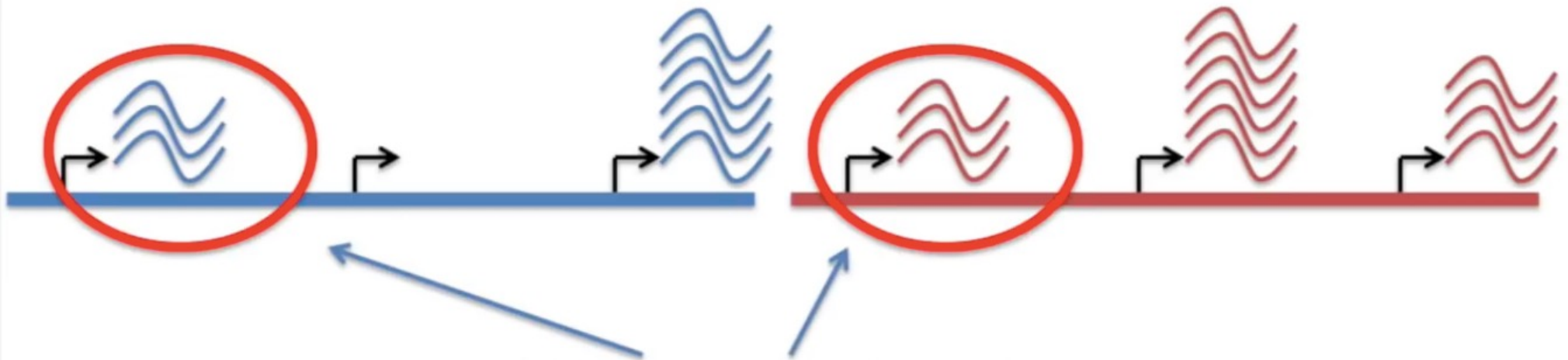
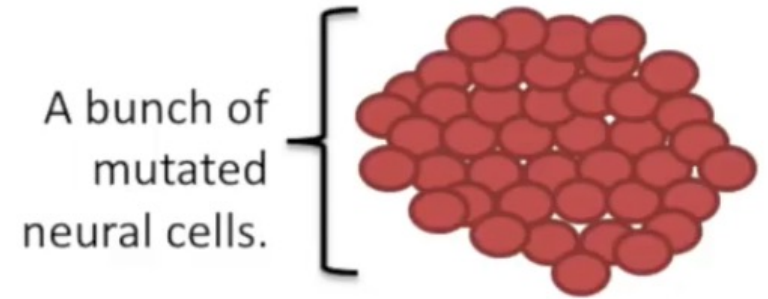
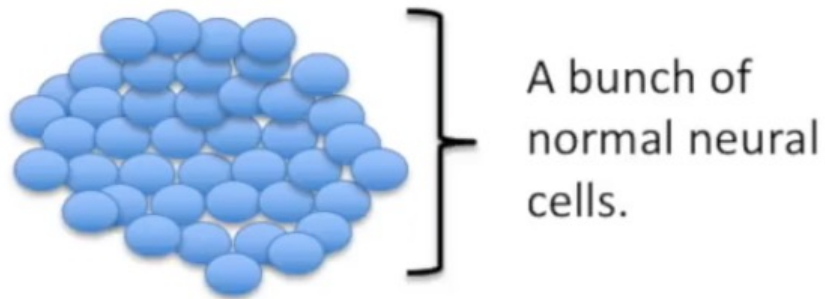
We can use RNA-seq to measure gene expression in normal cells...



... then use it to measure gene expression in mutated cells...

● = a normal neural cell

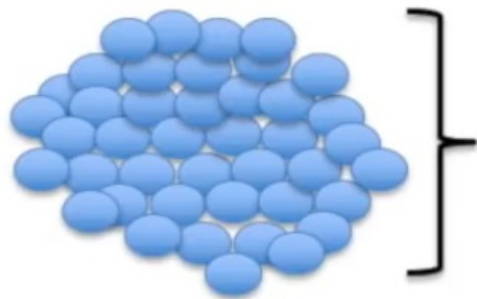
● = a mutated neural cell



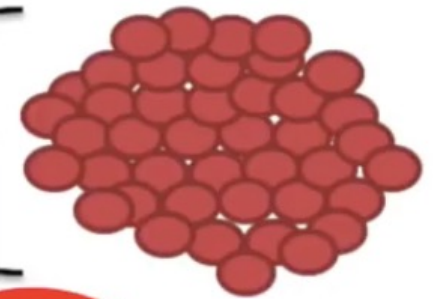
Gene1: No difference between normal and mutated cells.

● = a normal neural cell

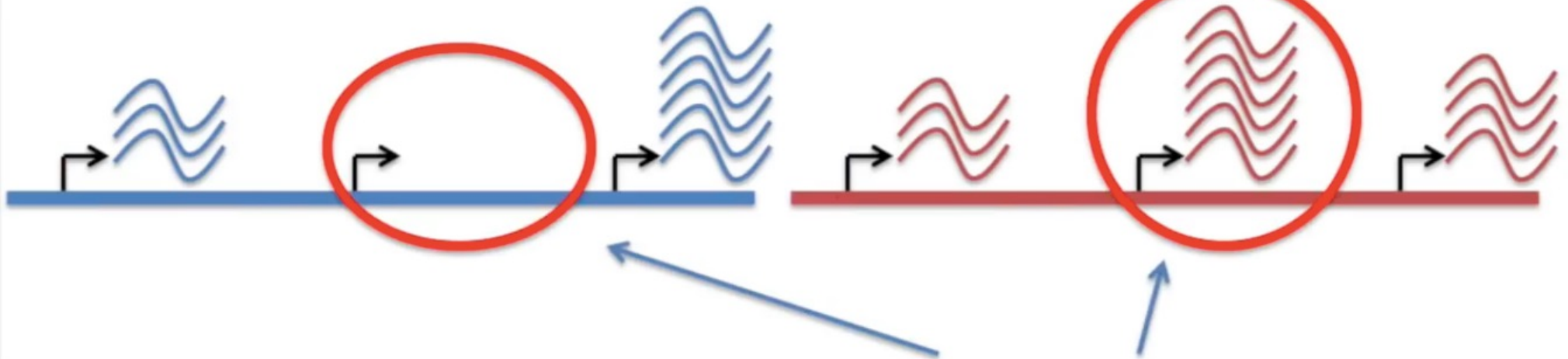
● = a mutated neural cell



A bunch of
normal neural
cells.



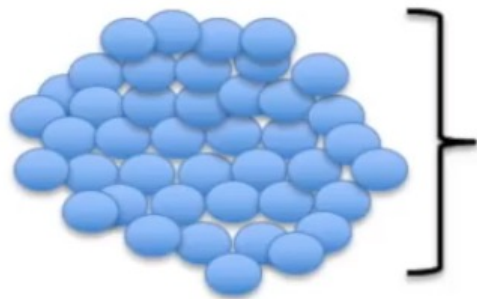
A bunch of
mutated
neural cells.



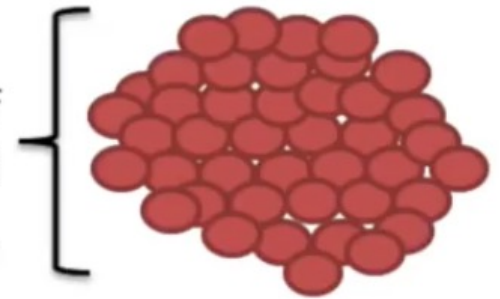
Gene2: A big difference between normal and mutated cells.

● = a normal neural cell

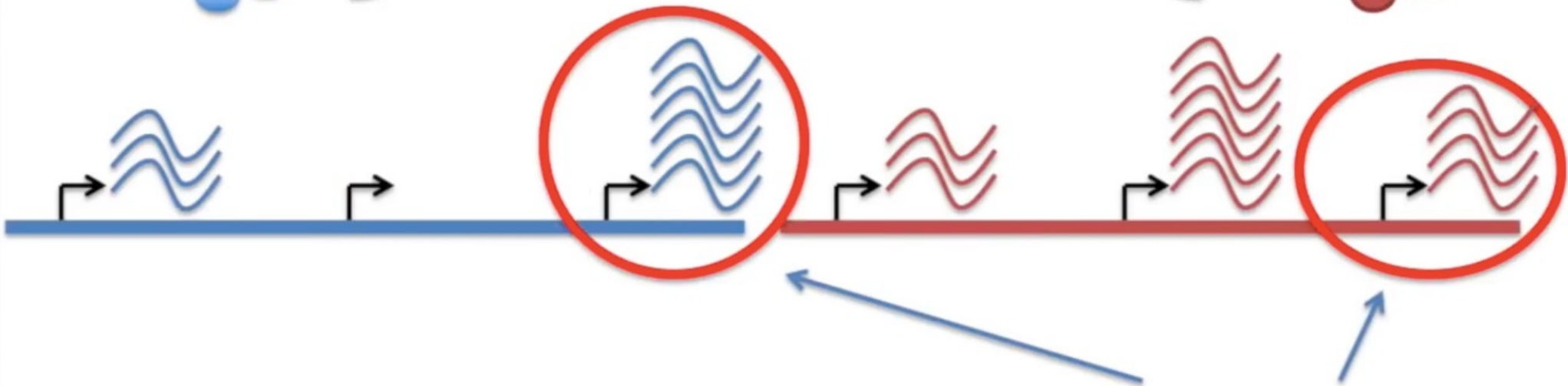
● = a mutated neural cell



A bunch of
normal neural
cells.



A bunch of
mutated
neural cells.



Gene3: A subtle difference between normal and mutated cells.

3 Main Steps for RNA-Seq:

- 1) Prepare a sequencing library
- 2) Sequence
- 3) Data analysis

Preparing an RNA-seq library

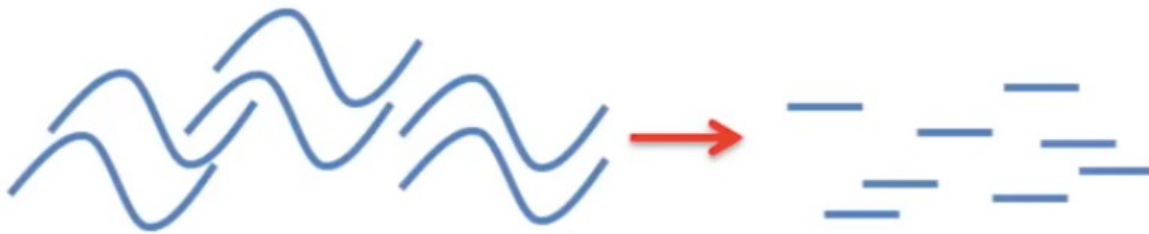
Step 1: Isolate the RNA



Preparing an RNA-seq library

Step 1: Isolate the RNA

Step 2: Break the RNA into small fragments.



We do this because RNA transcripts can be thousands of bases long, but the sequencing machine can only sequence short (200-300 bp) fragments

Preparing an RNA-seq library

Step 1: Isolate the RNA

Step 2: Break the RNA into small fragments.

Step 3: Convert the RNA fragments into double stranded DNA.



Double stranded DNA is more stable than RNA and can be easily amplified and modified. This leads us to the next step...

Preparing an RNA-seq library

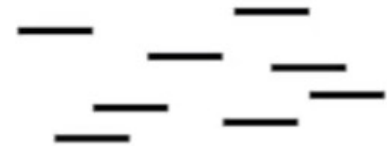
Step 1: Isolate the RNA



Step 2: Break the RNA into small fragments.



Step 3: Convert the RNA fragments into double stranded DNA.



Step 4: Add sequencing adaptors.



Preparing an RNA-seq library

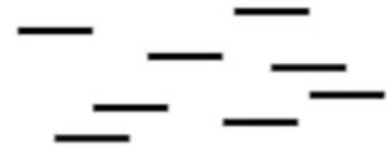
Step 1: Isolate the RNA



Step 2: Break the RNA into small fragments.



Step 3: Convert the RNA fragments into double stranded DNA.



Step 4: Add sequencing adaptors.



The adaptors do two things:

- 1) Allow the sequencing machine to recognize the fragments.
- 2) Allow you to sequence different samples at the same time, since different samples can use different adaptors.

Preparing an RNA-seq library

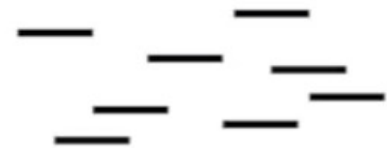
Step 1: Isolate the RNA



Step 2: Break the RNA into small fragments.



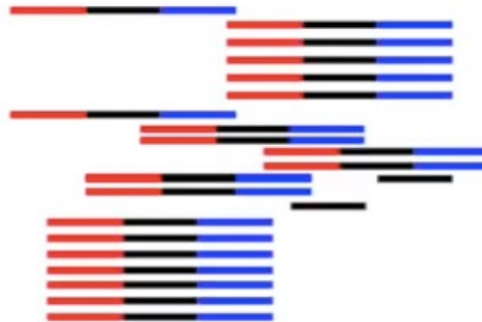
Step 3: Convert the RNA fragments into double stranded DNA.



Step 4: Add sequencing adaptors.



Step 5: PCR amplify.



Preparing an RNA-seq library

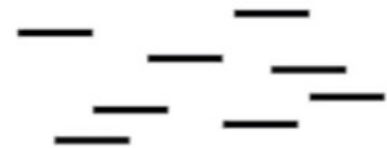
Step 1: Isolate the RNA



Step 2: Break the RNA into small fragments.



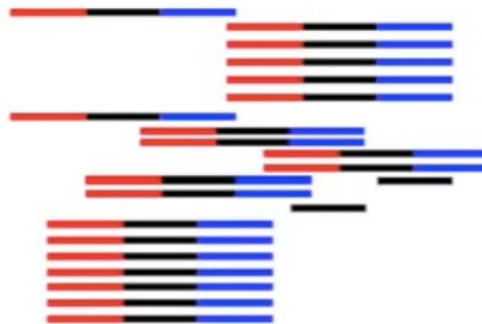
Step 3: Convert the RNA fragments into double stranded DNA.



Step 4: Add sequencing adaptors.



Step 5: PCR amplify.

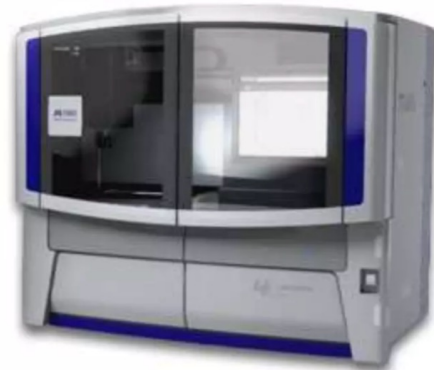


Step 6: QC

- 1) Verify library concentration
- 2) Verify library fragment lengths

Secuenciación de Siguiete Generación

Next-generation sequencing (NGS)



Minion



MinION Mk1: portable, real time biological analyses

MinION

Formato de archivo .fastq

```
@NS500177:196:HFTTAFXX:1:11101:10916:1458 2:N:0:CGCGGCTG
ACACGACGATGAGGTGACAGTCACGGAGGATAAGATCAATGCCCTCATTAAGCAGCCGGTGTAA
+
AAAAAEEEEEEEEEEEEEE//AEEEEEEEEEEEEEEEEEE/EE/<<EE/AEEEAEE///EEEEAEAEAEA<
```

Each sequencing “read” consists of 4
lines of data.

A typical sequence run with 400,000,000 reads
will generate a file containing 1.6 billion lines
of data!!!

Gene	Sample1	Sample2	Sample3...
A1BG	30	5	13...
A1BG-AS1	24	10	18...
A1CF	0	0	0...
A2M	5	9	7...
A2M-AS1	3563	5771	4123...
A2ML1	13	8	7...
...	...	...	...

If this were a Single Cell RNA-seq experiment, we would have 20,000 rows (genes) by 800+ columns (samples), giving us at least 16 million values to keep track of

1876 b

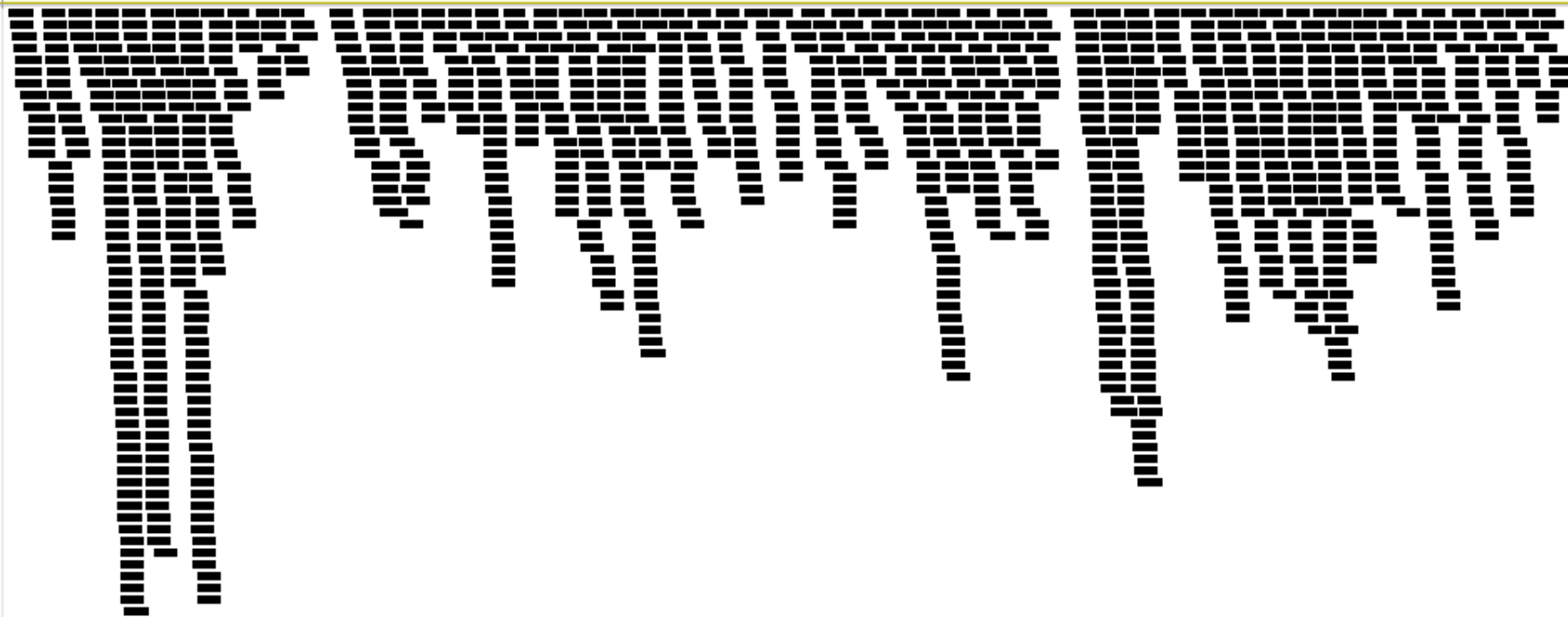
4491 b

3213 b

LOC115979980 gene

LOC115980031 gene

LOC115981243 gene



251 lecturas

404 lecturas

364 lecturas

1876 b

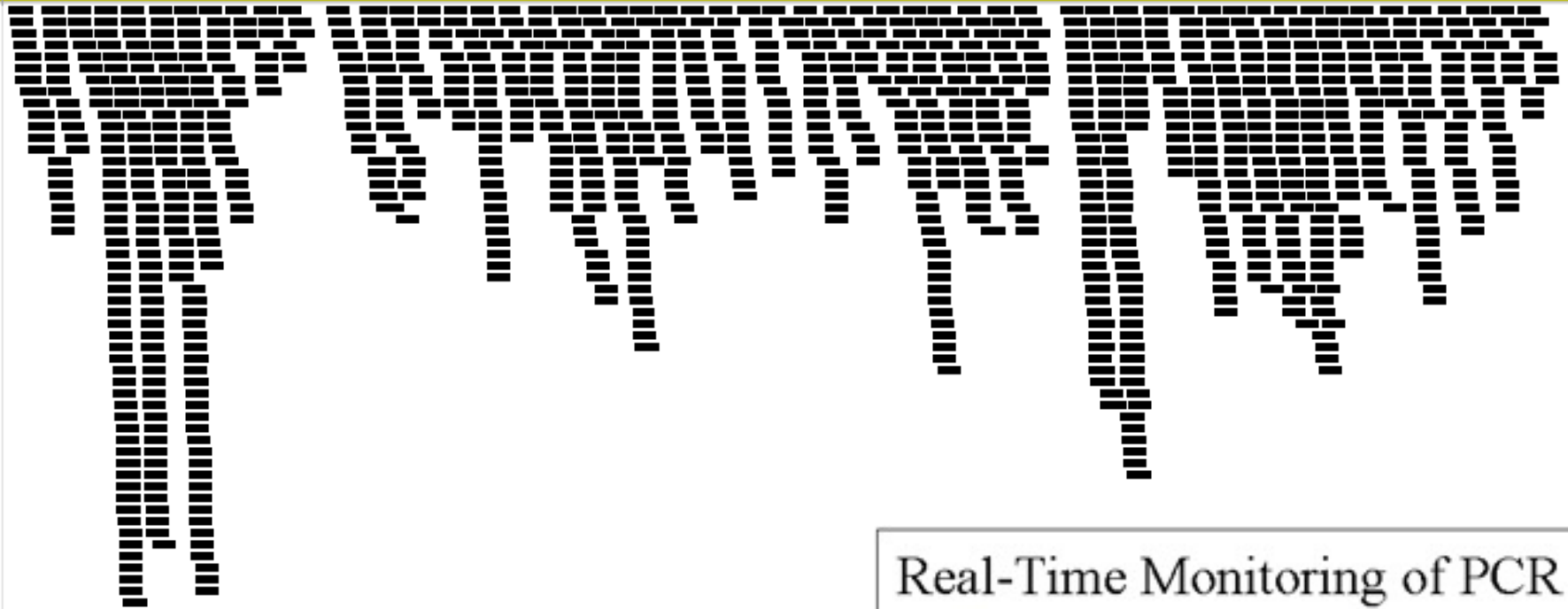
4491 b

3213 b

LOC115979980 gene

LOC115980031 gene

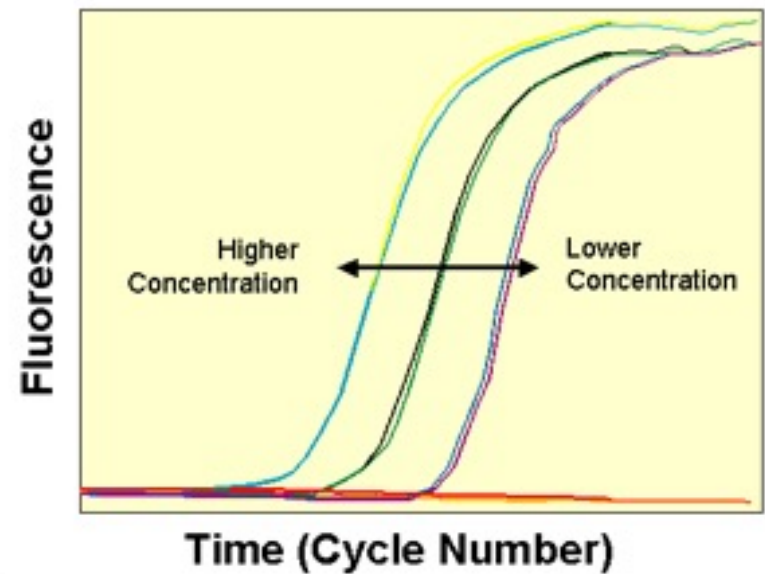
LOC115981243 gene



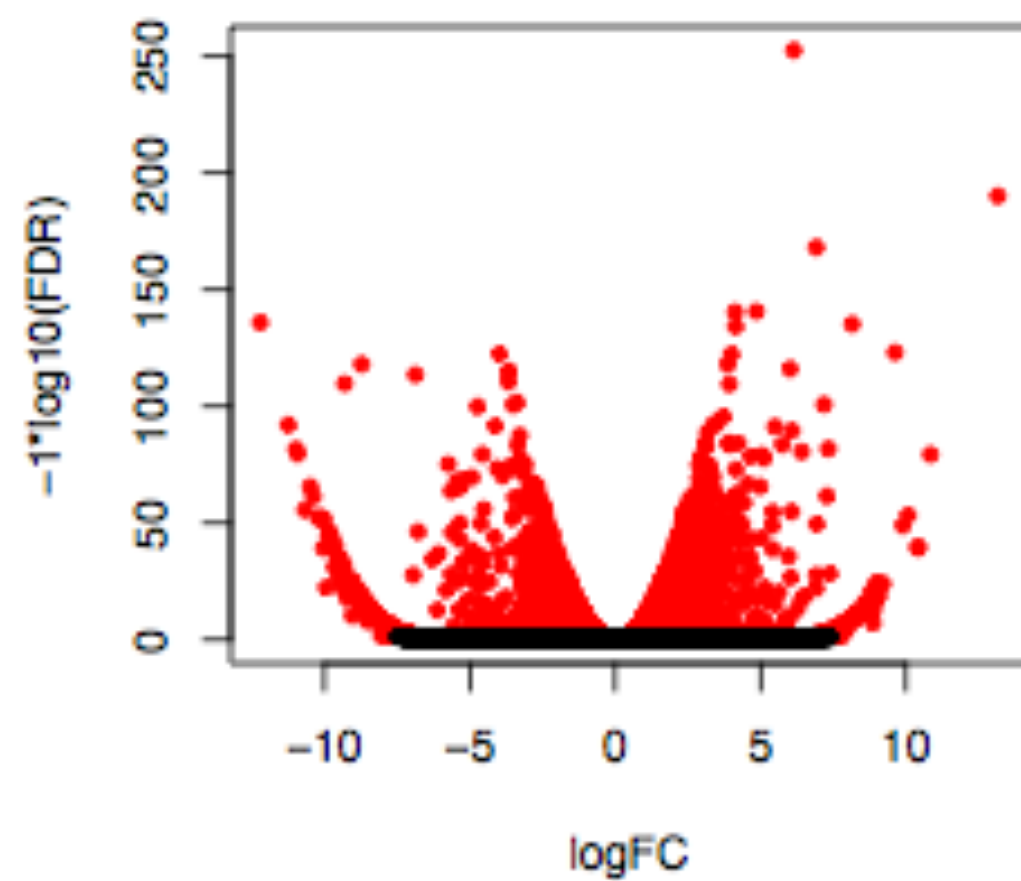
251 lecturas

404 lecturas

Real-Time Monitoring of PCR



Volcano plot



HEAT MAP

